

NORDIC ANALYTICS

Round 2 Technical Case Study

Fund Intelligence Dashboard Challenge

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Candidate Name	Role
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About this case

This case study is the second stage of the Nordic Analytics hiring process. It is a take-home challenge designed to be completed within 3–4 hours. You have 48 hours from receiving this document to submit your work. There are two versions of this case — one for the Frontend Engineer role and one for the Full Stack Tech Lead Intern role. Read the version that matches the role you are applying for. Both versions use the same mock dataset provided below.

What we are looking for

We are not looking for perfection. We are looking for your decision-making, your code quality, your ability to communicate technical choices, and your instincts about what matters. A clean, thoughtful partial solution with a good README is better than a rushed complete one. Show us how you think.

MOCK DATA

Provided to all candidates — use this dataset for your implementation

The following JSON dataset represents three funds managed by Nordic Analytics clients. This is simplified mock data — use it as-is. Do not spend time on realistic data generation.

```
// funds.json — Nordic Analytics Mock Dataset
{
  "funds": [
    {
      "id": "fund-001",
      "name": "Nordvest Capital Fund III",
      "type": "Private Equity",
      "vintage": 2019,
      "totalCommitments": 250000000,
      "metrics": {
        "irr": 18.4,
        "tvpi": 1.72,
        "dpi": 0.48,
        "rvpi": 1.24,
        "nav": 182000000
      },
      "navHistory": [
        {"month": "2024-01", "nav": 142000000},
        {"month": "2024-02", "nav": 147000000},
        {"month": "2024-03", "nav": 151000000},
        {"month": "2024-04", "nav": 154000000},
        {"month": "2024-05", "nav": 158000000},
        {"month": "2024-06", "nav": 161000000},
        {"month": "2024-07", "nav": 165000000},
        {"month": "2024-08", "nav": 168000000},
        {"month": "2024-09", "nav": 171000000},
        {"month": "2024-10", "nav": 174000000},
        {"month": "2024-11", "nav": 178000000},
        {"month": "2024-12", "nav": 182000000}
      ],
      "portfolioCompanies": [
        {
          "id": "pc-001", "name": "Lumio AS", "sector": "SaaS", "country": "Norway",
          "revenue": 12400000, "ebitda": 3100000, "ebitdaMargin": 25.0, "status":
"Active",
          "investmentDate": "2020-03-15", "investedCapital": 8000000,
          "currentValue": 22000000,
          "flags": []},
        {
          "id": "pc-002", "name": "FleetOS GmbH", "sector": "Logistics Tech",
          "country": "Germany",
          "revenue": 7800000, "ebitda": 780000, "ebitdaMargin": 10.0, "status":
"Active",
          "investmentDate": "2021-06-01", "investedCapital": 5000000,
          "currentValue": 6200000,
          "flags": ["watch"]},
        {
          "id": "pc-003", "name": "BioStride Ltd", "sector": "HealthTech",
          "country": "UK",
          "revenue": 3200000, "ebitda": -480000, "ebitdaMargin": -15.0, "status":
"Active",
          "investmentDate": "2022-09-10", "investedCapital": 6000000,
          "currentValue": 5100000,
          "flags": ["at-risk"]}
      ]
    },
    {
      "id": "fund-002",
```

```

    "name": "Baltic Ventures Fund I",
    "type": "Venture Capital",
    "vintage": 2021,
    "totalCommitments": 80000000,
    "metrics": {"irr": 31.2, "tvpi": 1.41, "dpi": 0.12, "rvpi": 1.29, "nav":
72000000},
    "navHistory": [
      {"month": "2024-01", "nav": 52000000}, {"month": "2024-02", "nav":
54000000},
      {"month": "2024-03", "nav": 55000000}, {"month": "2024-04", "nav":
57000000},
      {"month": "2024-05", "nav": 59000000}, {"month": "2024-06", "nav":
61000000},
      {"month": "2024-07", "nav": 63000000}, {"month": "2024-08", "nav":
65000000},
      {"month": "2024-09", "nav": 67000000}, {"month": "2024-10", "nav":
69000000},
      {"month": "2024-11", "nav": 71000000}, {"month": "2024-12", "nav":
72000000}
    ],
    "portfolioCompanies": [
      {"id": "pc-004", "name": "Kreditt.io", "sector": "Fintech", "country":
"Finland",
        "revenue": 4200000, "ebitda": 630000, "ebitdaMargin": 15.0, "status":
"Active",
        "investmentDate": "2021-11-20", "investedCapital": 3000000,
        "currentValue": 9800000,
        "flags": []},
      {"id": "pc-005", "name": "CloudTrace AB", "sector": "Dev Tools", "country":
"Sweden",
        "revenue": 2100000, "ebitda": -420000, "ebitdaMargin": -20.0, "status":
"Active",
        "investmentDate": "2022-04-05", "investedCapital": 2500000,
        "currentValue": 3200000,
        "flags": ["watch"]}
    ]
  },
  {
    "id": "fund-003",
    "name": "Nordic Growth Equity II",
    "type": "Growth Equity",
    "vintage": 2018,
    "totalCommitments": 400000000,
    "metrics": {"irr": 14.1, "tvpi": 2.08, "dpi": 1.45, "rvpi": 0.63, "nav":
98000000},
    "navHistory": [
      {"month": "2024-01", "nav": 118000000}, {"month": "2024-02", "nav":
115000000},
      {"month": "2024-03", "nav": 112000000}, {"month": "2024-04", "nav":
110000000},
      {"month": "2024-05", "nav": 108000000}, {"month": "2024-06", "nav":
106000000},
      {"month": "2024-07", "nav": 104000000}, {"month": "2024-08", "nav":
102000000},
      {"month": "2024-09", "nav": 101000000}, {"month": "2024-10", "nav":
100000000},
      {"month": "2024-11", "nav": 99000000}, {"month": "2024-12", "nav":
98000000}
    ],
    "portfolioCompanies": [
      {"id": "pc-006", "name": "Borealis Energy", "sector": "CleanTech",
"country": "Denmark",
        "revenue": 31000000, "ebitda": 8680000, "ebitdaMargin": 28.0, "status":
"Active",
        "investmentDate": "2019-02-28", "investedCapital": 25000000,
        "currentValue": 61000000,

```

```
      "flags": [],  
      {"id": "pc-007", "name": "Timberwolf Logistics", "sector": "Logistics",  
"country": "Norway",  
      "revenue": 18500000, "ebitda": 1480000, "ebitdaMargin": 8.0, "status":  
"Active",  
      "investmentDate": "2019-08-12", "investedCapital": 15000000,  
"currentValue": 17200000,  
      "flags": ["watch"]}  
    ]  
  }  
]  
}
```

TRACK A — FRONTEND ENGINEER

Complete this section if applying for the Frontend Engineer role

Before you start

This track is for Frontend Engineer candidates. You should not implement a backend server or database — all data should come from the provided funds.json file imported directly into your React app. Focus on code quality, component design, data visualization, and UX clarity.

Your Task

Build a Portfolio Performance Dashboard as a single-page React application (TypeScript required). The dashboard is used by a fund manager to monitor the performance of their fund portfolio. It should look and feel like real institutional software — not a tutorial project.

Required Deliverables

1. Fund Selector

A tab or dropdown that allows the user to switch between the three funds in the dataset. The dashboard should update reactively when a fund is selected.

2. KPI Summary Cards

Display the key metrics for the selected fund: IRR (%), TVPI (x), DPI (x), RVPI (x), and current NAV (formatted as currency). Cards should be clearly labeled and easy to read at a glance.

3. NAV Performance Chart

A line chart showing the 12-month NAV history for the selected fund using any charting library (Recharts, Chart.js, or D3.js are all acceptable). Axis labels and tooltips must be readable and correctly formatted. The chart must update when the user switches funds.

4. Portfolio Company Table

A sortable data table displaying all portfolio companies for the selected fund. Required columns: Company Name, Sector, Country, Revenue, EBITDA, EBITDA Margin (%), Current Value, and Status. Rows flagged as 'watch' should be visually distinguished (e.g., amber row or badge). Rows flagged as 'at-risk' should be clearly highlighted in red or with a warning indicator.

5. README

A written README.md that explains: (a) how to run the project locally, (b) your key technical decisions and why you made them, (c) what you would improve or add given more time, and (d) any known limitations.

Bonus (Optional)

- Multi-fund NAV overlay: allow the user to toggle multiple funds on the same chart for comparison
- Column sorting: clicking a column header sorts the portfolio table by that column

- Threshold alert: display a banner if any portfolio company has a negative EBITDA margin
- Responsive layout: the dashboard is usable on smaller screen widths

Technical Constraints

- React with TypeScript is required. Use of Next.js is optional but welcome.
- Tailwind CSS is strongly preferred for styling. Other utility frameworks are acceptable.
- No backend server. Import funds.json directly. You may use a static JSON import or a mock API using json-server if you prefer.
- Any chart library is acceptable — Recharts, Chart.js, D3.js, Nivo, etc.
- No authentication required for Track A.

Submission

How to submit

Send a link to a GitHub (or GitLab) repository. The repo must include your README.md and all source code. If the project requires any env setup, document it clearly. We will clone and run your project. If it does not run, we will reach out once to clarify — after that we will assess what we can see.